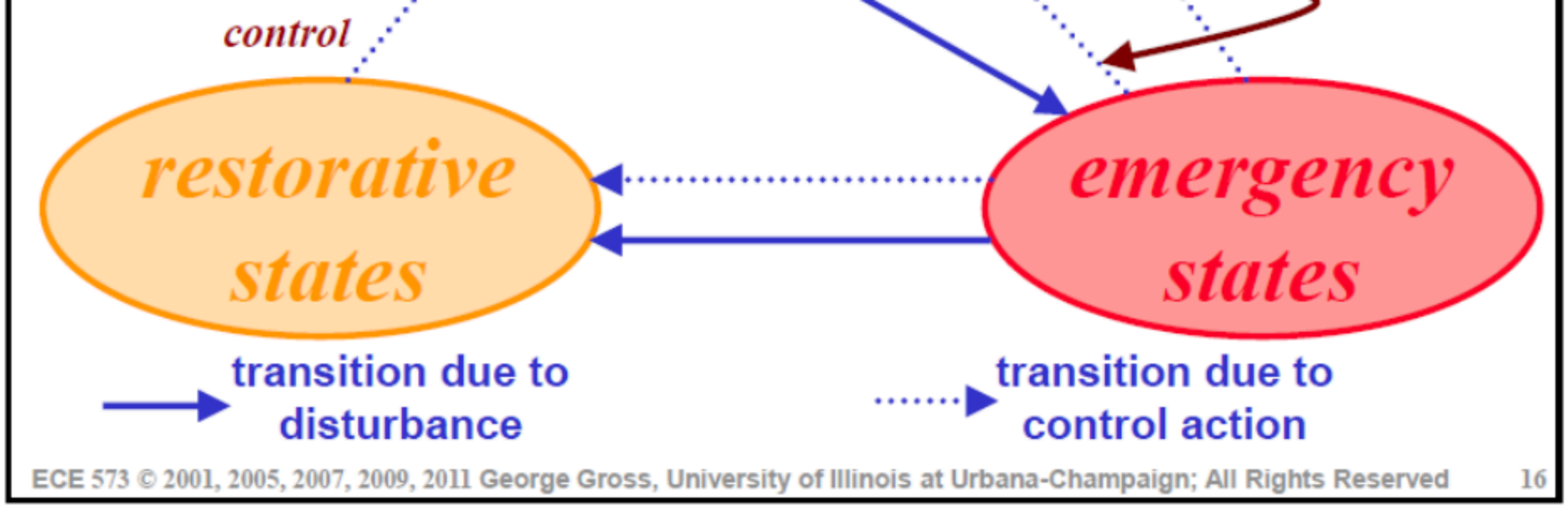

FIGURE 2: POWER SYSTEM SECURITY STUDY FRAMEWORK

mplementing the security framework into the Energy Management System (EMS) can provide the perators online security analysis functionality to closely monitor, assess and control system ecurity.

2.2 PREVENTIVE MARKET OPTMIZATION

ection 2.1 discussed the security framework that is applicable to system operations. In this ection, we will focus on the market aspect of helping power system security. As discussed in ection 2.1, the preferred power system operating state is the secure state. In electricity markets, e market solution typically tries to operate the system under secure state. In order to achieve N-1 ecurity, the market optimization, typically an optimal power flow (OPF) program or a unit ommitment (UC) program, will consider every contingency in the contingency list, and include onstraints of the immediate post contingency system conditions. The decision variables are the